Life Cycle Assessment and Life Cycle Costing of a Combined Pyrolysis and Anaerobic Digestion Plant

Extension of the Cooperative Design Project

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Abstract

This report adds a life cycle assessment (LCA) and a life cycle cost calculation (LCC) to the combined pyrolysis and anaerobic digestion plant that my group designed in the cooperative design project at TUM Campus Straubing. The design report had neither. I modelled the plant for 1 t of feedstock in openLCA and checked the result against a separate Excel model, which also holds the cost calculation.

The base case causes about 36 kg CO2e per tonne of feedstock. If the heat from the CHP engine is sold, the result drops to about -5 kg CO2e per tonne. At the 2025 spot price for electricity the net cost is about EUR 17 per tonne and the plant does not pay back (NPV about -EUR 44 million). With a tariff from the EEG biomass tender the NPV is positive (about +EUR 15 million), but that figure is an upper bound. Some inputs are still my own assumptions. I list them, with their effect, in section 9.

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1.0 Introduction

1.1 Background

In the design project we planned a biogas plant with a capacity of 1,500 m³ CH4 per hour. It is fed with cow manure and energy crops, and a pyrolysis unit that treats green waste sits next to it. Biochar, bio-oil and syngas from the pyrolysis go to the CHP engine together with the biogas. The report covered the process design and a rough economic estimate. It said nothing about environmental impact, and when I went back to the economic part, I found that it did not hold together.

I wanted to close that gap with a small piece of work: an LCA and an LCC that run on the same inventory, so that no flow is counted twice and a change in one input shows up in both results.

1.2 Aim of this work

The work has four aims:

- Build a life cycle assessment of the plant in openLCA for 1 t of feedstock.
- Check the result with an independent calculation in Excel.
- Add a life cycle cost calculation with sourced inputs to the same Excel model.
- State clearly which inputs are sourced and which are assumptions, and how much they matter.

2.0 Problems in the design report

Before building anything, I checked the numbers of the design report. Four of them were wrong or inconsistent and would have gone straight into the LCA and the cost model. Table 1 shows what they are and what I did about them.

Issue in the design report	Why it matters	How this work handles it
Pyrolysis capacity of about 17.5 million t per year (a unit mix-up between 0.125 m³/t and a 12.5 % yield)	Reactor volume and flow rates that follow from it are meaningless	Pyrolysis is now 10 % of the feedstock mass. This is my own assumption
Fermenter volume of 900,000 m³ (methane flow times time, not substrate volume)	Wrong reactor size and cost	Not used. The model needs no reactor volume
Two different methane yields for crops (100 and 250 m³/t)	Changes the energy output by a factor of 2.5	One value: 100 m³ CH4 per t of crops, plus 20 m³ per t of manure
Economics do not fit the plant size (about 57,600 MWh of electricity per year but only $200k revenue; CAPEX of $4.7M for about 6.5 MWel)	An NPV on these numbers would be wrong	All cost inputs are replaced by sourced values (section 8)

Table 1: Problems found in the design report.

I kept the inputs that were consistent: 1,500 m³ CH4/h, 8,760 operating hours per year, 197,100 t/yr of manure and 91,980 t/yr of crops, the CHP efficiencies (40 % electrical, 40 % thermal) and the pyrolysis product split of 30 % char, 40 % bio-oil and 30 % syngas.

3.0 Goal and scope

The functional unit is 1 t of feedstock treated. A per-tonne basis was the safer choice, since the plant-scale numbers in the design report could not be trusted, and it also gives the cost per tonne directly. Table 2 lists the main choices.

Choice	What I did	Reason
Functional unit	1 t of feedstock	Safer than the plant-scale numbers. Gives cost per tonne
Feedstock per tonne	0.614 t cow manure, 0.286 t crops (maize, wheat and grass in equal parts) and 0.10 t green waste for the pyrolysis unit	Manure and crop shares come from the design report. The 10 % for pyrolysis is my assumption
Boundary	Crops up to the farm gate, AD and pyrolysis units, CHP engine, grid electricity and direct emissions	Plant construction is left out

Choice	What I did	Reason
Method	IPCC 2021 (AR6), GWP100, in kg CO2e	Biogenic CO2 is not counted
Manure and green waste	Zero burden	They are residues from other activities (cut-off rule)
Credits	Only exported electricity, as avoided grid power	I cannot show that anyone would use the heat, biochar or digestate. Heat is a scenario
Software	openLCA 2.5.0 with Agribalyse 3.2, and Excel	Excel is the cross-check and holds the costs

Table 2: Goal and scope.

4.0 Data sources

I did not have access to the ecoinvent database, so I used the free Agribalyse 3.2 database (French agriculture and food) for the crops and a published national factor for electricity. Table 3 lists each source and its limit, so that a reader can judge the result.

Part of the model	Source	Limit
Maize, wheat, grass	Agribalyse 3.2 core, conventional, at farm gate	French average or regional data used as a proxy for Germany
Grid electricity and credit	German grid mix 2025, 344 g CO2/kWh (Umweltbundesamt)	CO2 only. The 2025 value is an estimate
Climate indicator	IPCC 2021 GWP100 from the openLCA LCIA methods	Biogenic CO2 not counted
Direct emissions	IPCC AR6: CH4 (non-fossil) = 27.0, N2O = 273; IPCC 2006 default of 1 % of applied nitrogen emitted as N2O-N	Methane slip and digestate nitrogen are uncertain
Heat scenario	Natural gas 240 g CO2e/kWh (GEG Anlage 9)	Boiler efficiency of 90 % is assumed

Table 3: Data sources and limits.

4.1 Two database results that I rejected

I compared every result with a realistic range before using it. Two results failed this check.

First, the ELCD 3.2 German grid mix gave 3.18 kg CO2e for 1 kWh (Figure 1). This is six to eight times higher than a realistic value for Germany, which is roughly 0.3 to 0.7 kg. I had set the calculation up correctly (1 kWh, target amount 1), so the data itself was the problem and I did not use it.

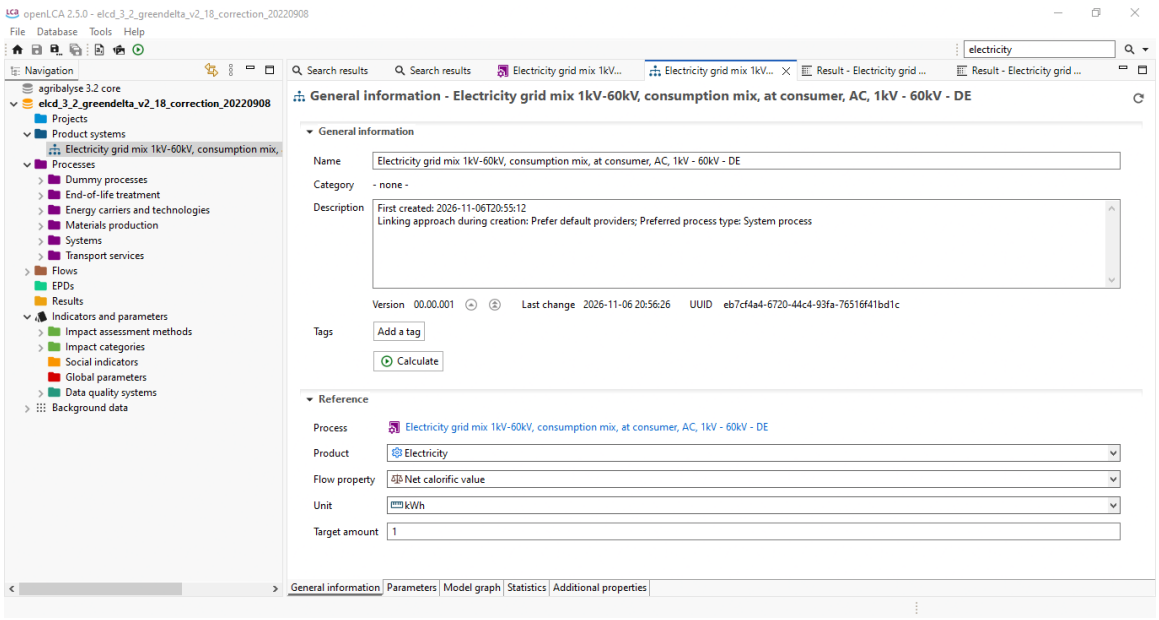


Figure 1: ELCD 3.2 German consumption mix, set up for 1 kWh. Result: 3.18 kg CO2e, rejected.

Second, the Agribalyse copy of the ecoinvent electricity process ({DE}, market for) gave 0 kg CO2e. Its inputs table is empty and it only lists 1 kWh as output (Figure 2). It is a placeholder, because the free Agribalyse version does not contain the ecoinvent background data.

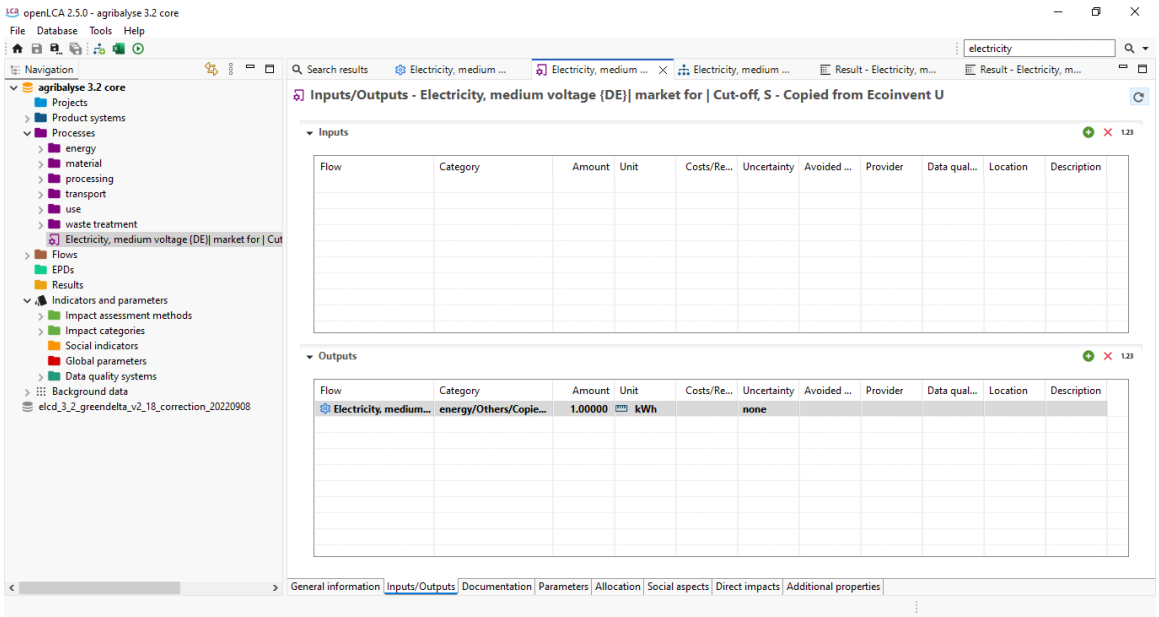


Figure 2: The Agribalyse copy of the ecoinvent electricity process: empty inputs table, only the 1 kWh output. Result: 0, rejected.

From this I took a simple rule. Before a number goes into the model, open the inputs and outputs tab and compare the result with a realistic range. A zero or an odd value usually means that the process is empty or wrong.

I replaced both with the UBA factor of 0.344 kg CO2 per kWh (Umweltbundesamt, 2025), entered as a small process of my own (section 6, step 4).

5.0 Inventory for 1 t of feedstock

The inventory in Table 4 is the one list that both the GWP and the cost calculation use, so nothing is counted twice. I entered the same amounts in openLCA.

Flow	Amount	GWP factor	GWP (kg CO2e)	Source of factor
Cattle manure (input)	0.614 t	0	0	Waste, cut-off rule
Silage maize (input)	95.45 kg	202.47 kg CO2e/t	+19.33	Agribalyse 3.2, farm gate
Soft wheat grain (input)	95.45 kg	391.88 kg CO2e/t	+37.41	Agribalyse 3.2, farm gate
Grass silage (input)	95.45 kg	189.53 kg CO2e/t	+18.09	Agribalyse 3.2, farm gate, NW France
Green waste to pyrolysis (input)	0.10 t	0	0	Waste, cut-off rule
Grid electricity, plant use (input)	25 kWh	0.344 kg CO2e/kWh	+8.60	UBA 2025
Methane slip, 1.9 % of 40.9 Nm³ CH4 (emission)	0.557 kg	27.0	+15.05	IPCC AR6; slip rate from DBFZ / IEA Task 37
N2O from digestate on the field (emission)	0.0764 kg	273	+20.85	IPCC AR6; 6.0 kg N per t digestate (LfL Bayern), 1 % EF
Electricity exported by the CHP (avoided grid electricity)	240.9 kWh	0.344 kg CO2e/kWh	-82.86	UBA 2025
Total, base case			**+36.46**	

Table 4: Inventory and GWP for 1 t of feedstock.

I did not credit the exported heat (156.6 kWh), the biochar (18 kg) or the digestate (0.81 t) in the base case.

5.1 How I worked out the energy

- **Anaerobic digestion:** 0.614 t manure x 20 + 0.286 t crops x 100 = 40.9 Nm³ CH4. At 10.97 kWh/Nm³ (39.5 MJ/Nm³) this is 449 kWh.
- **Pyrolysis:** 100 kg of green waste at 40 % moisture leaves 60 kg dry matter. 30 % of it becomes 18 kg of biochar, 40 % bio-oil and 30 % syngas. With heating values of 17 MJ/kg for the bio-oil and 8 MJ/kg for the syngas they carry 153 kWh.

- **CHP:** (449 + 153) kWh x 40 % = 240.9 kWh of electricity. The heat is 40 % x (1 - 35 % used on site) = 156.6 kWh, which is exported.

5.2 Choice of the maize process

My first maize value (235.4 kg CO2e/t) came from the "at feeder" process, which includes about 16 % feeding losses that are specific to cattle. I changed to the farm-gate process (202.5 kg CO2e/t), so that maize, wheat and grass are all taken at the farm gate and the digester is not charged for cattle feeding losses.

6.0 Modelling in openLCA

The model has two layers. The first is a unit GWP for each background flow. The second is one foreground process that represents the plant, per 1 t of feedstock.

Step 1: Importing the databases and the method

I imported Agribalyse 3.2 core with Databases > Import a database, and ELCD 3.2 as a separate database, so the flow lists do not get mixed up (Figure 3). Under Impact assessment methods I checked that IPCC 2021 was available (Figure 4) and used GWP100 for all calculations.

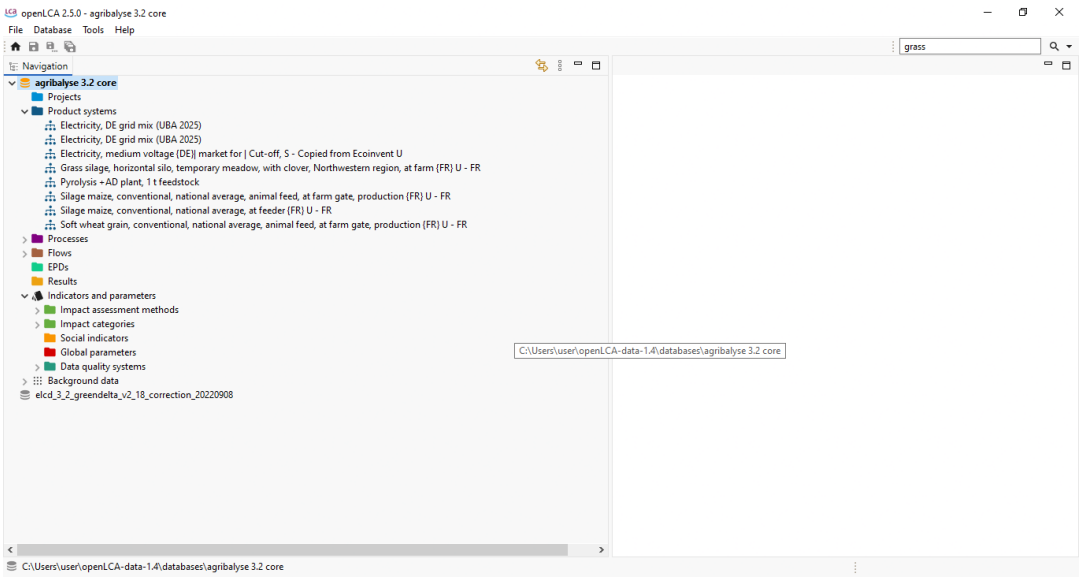


Figure 3: Both databases in the Navigation panel, with the product systems created during the work.

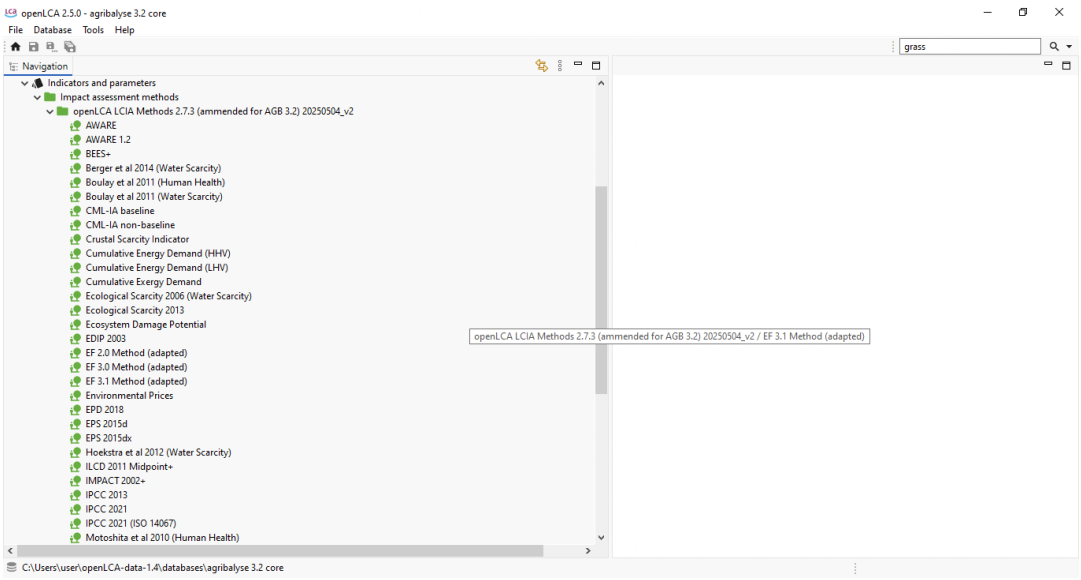


Figure 4: Impact assessment methods after the import (openLCA LCIA Methods 2.7.3). IPCC 2021 and IPCC 2021 (ISO 14067) are present.

Step 2: Finding and checking the crop processes

I searched the processes by keyword and only used the ones with a purple process icon, since the blue gear icons are flows and cannot be calculated. I skipped names with "organic", "scenario", "seed" or "stored and transported" and preferred "conventional, national average". Each process was opened in the inputs and outputs tab to check that it was not empty (Figure 5).

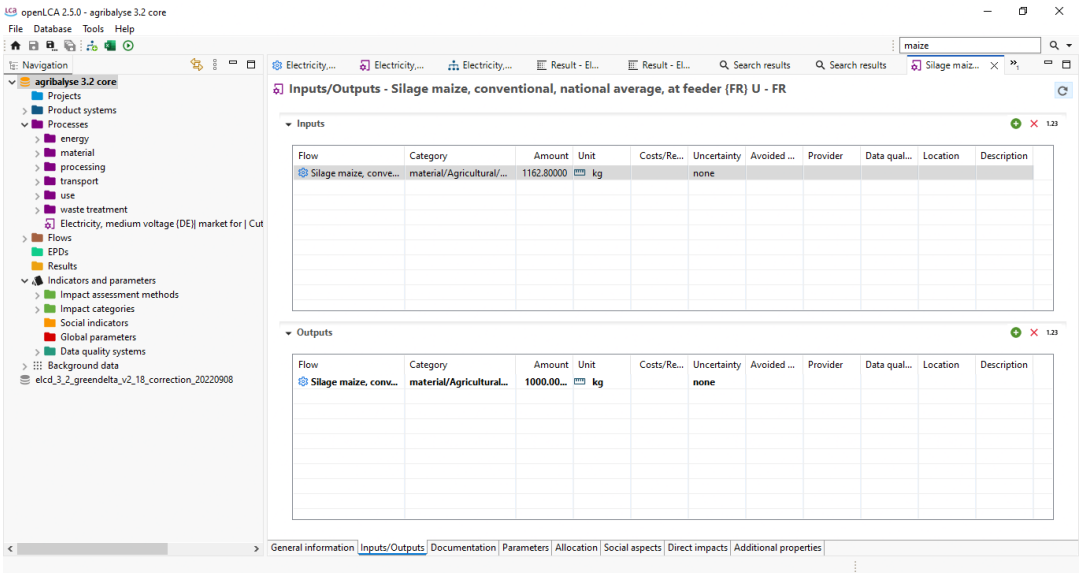


Figure 5: A populated Agribalyse process (silage maize, at feeder): 1,000 kg output, 1,162.8 kg upstream maize input.

Step 3: Getting the GWP of each crop

For each process I used Create product system with a target amount of 1,000 kg, so that the result is per tonne, and calculated it with IPCC 2021 GWP100. I got 202.47 kg CO2e per tonne for silage maize, 391.88 for soft wheat grain and 189.53 for grass silage. Figure 6 shows the wheat result.

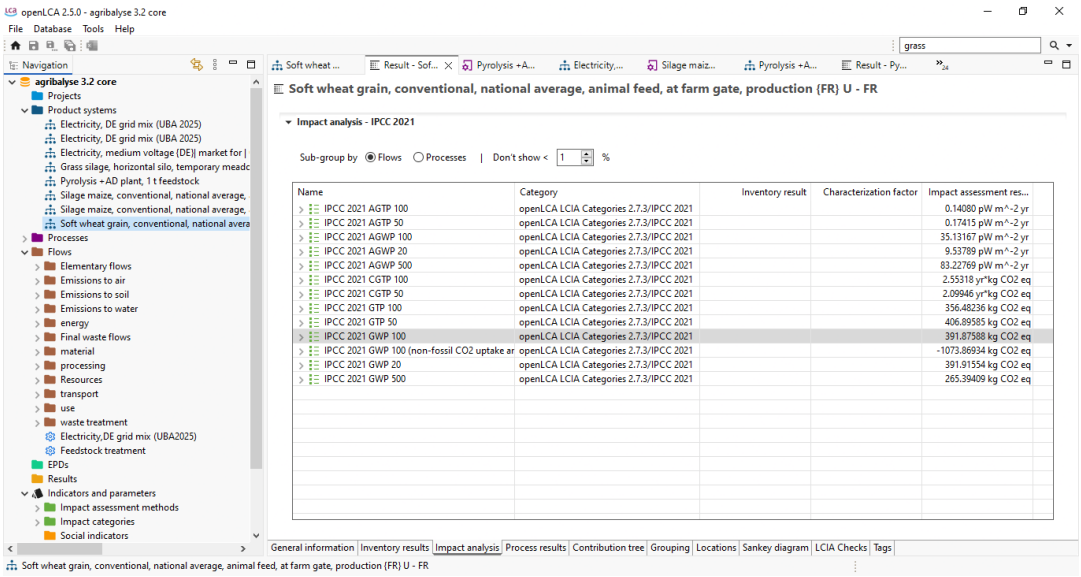


Figure 6: Impact analysis for 1,000 kg of soft wheat grain: IPCC 2021 GWP 100 = 391.88 kg CO2 eq.

Step 4: Making my own electricity process

No database process passed the check, so I made a small one by hand. It has a new product flow "Electricity, DE grid mix (UBA 2025)" in kWh and a process that outputs 1 kWh together with 0.344 kg of the elementary flow Carbon dioxide, fossil, to air (Figure 7).

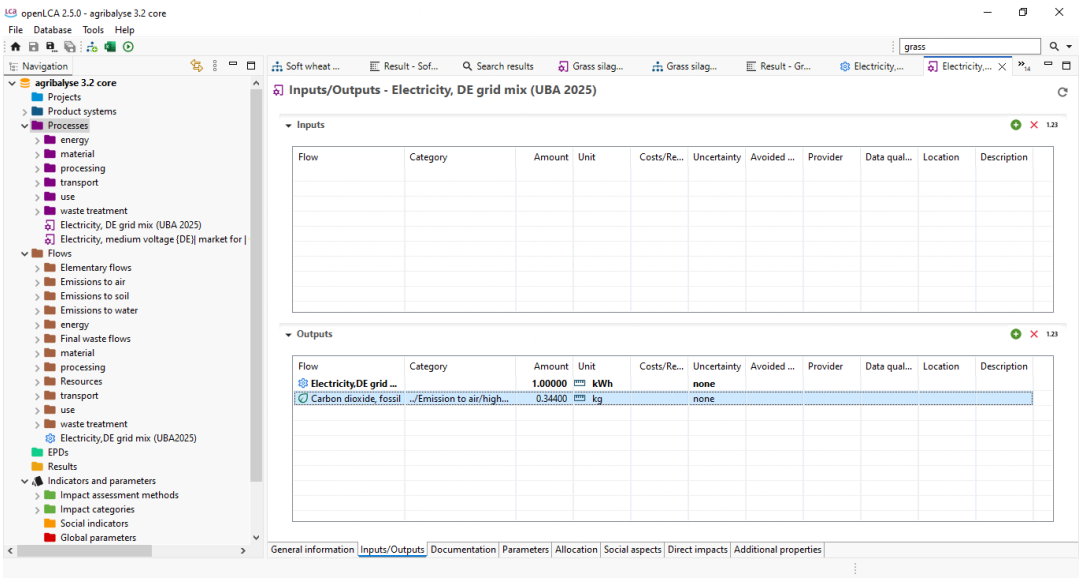


Figure 7: The custom electricity process: 1 kWh output and 0.344 kg CO2 (fossil) emission.

Step 5: Building the plant process

I created a new process, "Pyrolysis + AD plant, 1 t feedstock", with the product flow "Feedstock treatment" (1 t) as its reference (Figure 8). The inputs link to the provider processes for maize, wheat, grass and electricity (Figure 9). The exported electricity is an output marked as avoided product. Methane (non-fossil) and nitrous oxide are entered as emissions to air (Figure 10).

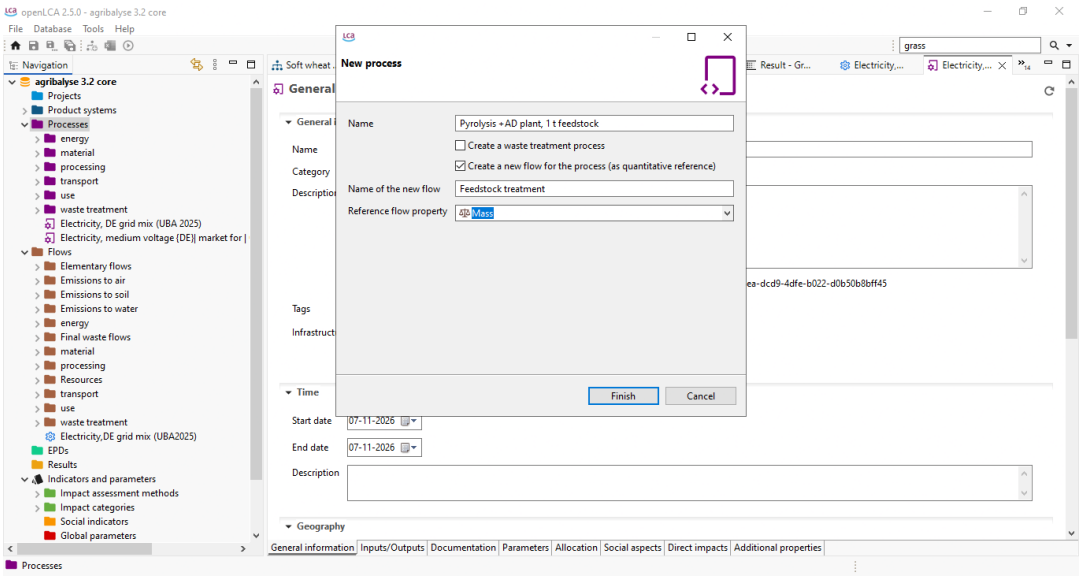


Figure 8: Creating the plant process with a new reference flow.

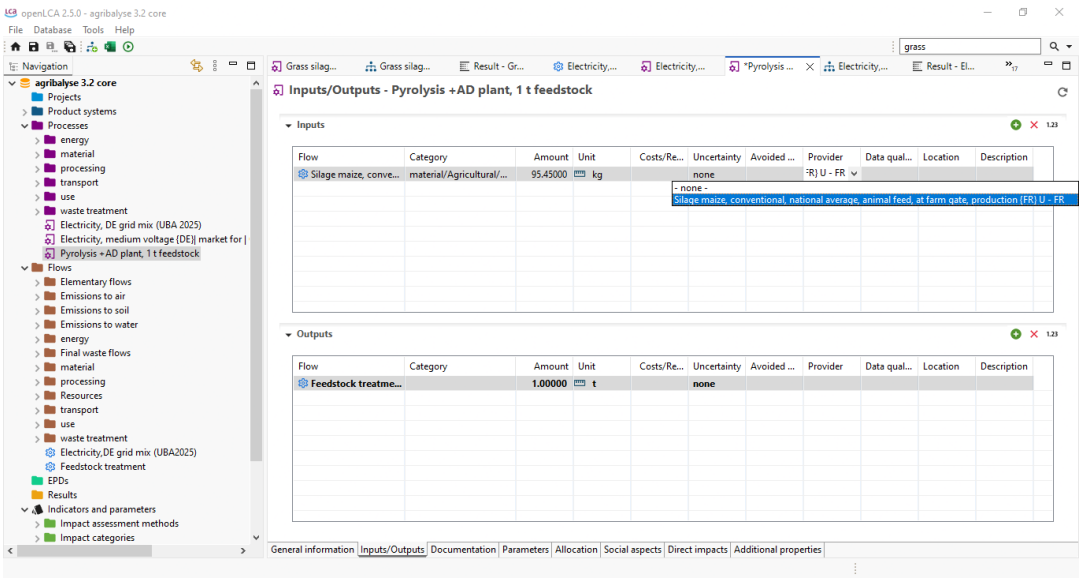


Figure 9: Linking an input to its provider process (silage maize, farm gate).

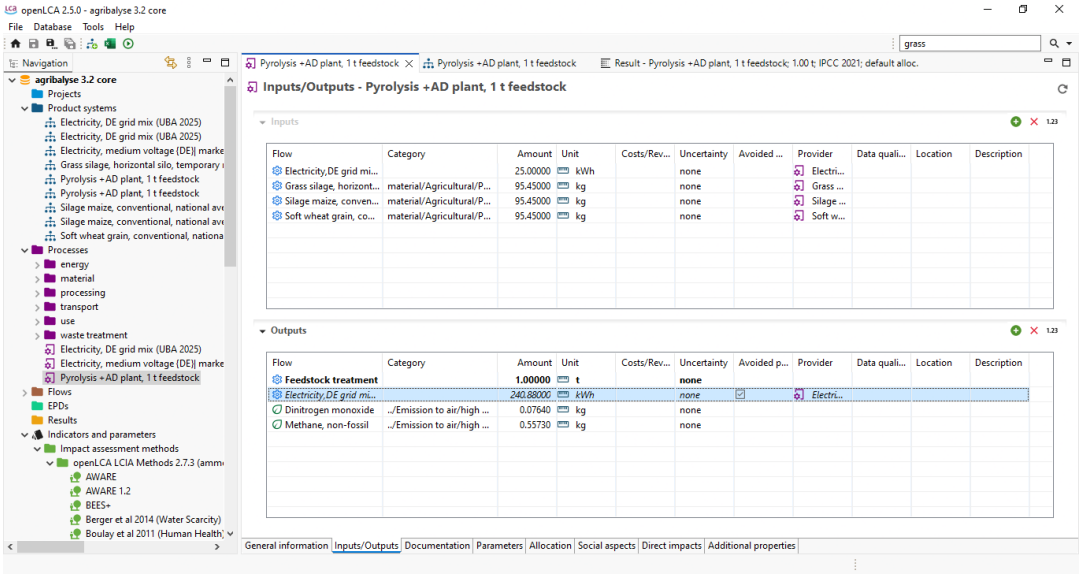


Figure 10: The finished plant process: four inputs with providers, the avoided electricity (240.88 kWh) and two direct emissions (0.0764 kg N2O, 0.5573 kg methane).

Step 6: Calculating the plant

I turned the process into a product system with a target amount of 1 t and calculated it with IPCC 2021 GWP100. My first run gave 23.265 kg CO2 eq. After I found sources for the methane slip (1.9 %), the digestate nitrogen (6.0 kg N per t) and the syngas heating value (8 MJ/kg), I changed three amounts: methane to 0.5573 kg, N2O to 0.0764 kg and the avoided electricity to 240.88 kWh. The final result is 36.463 kg CO2 eq per t of feedstock (Figure 11).

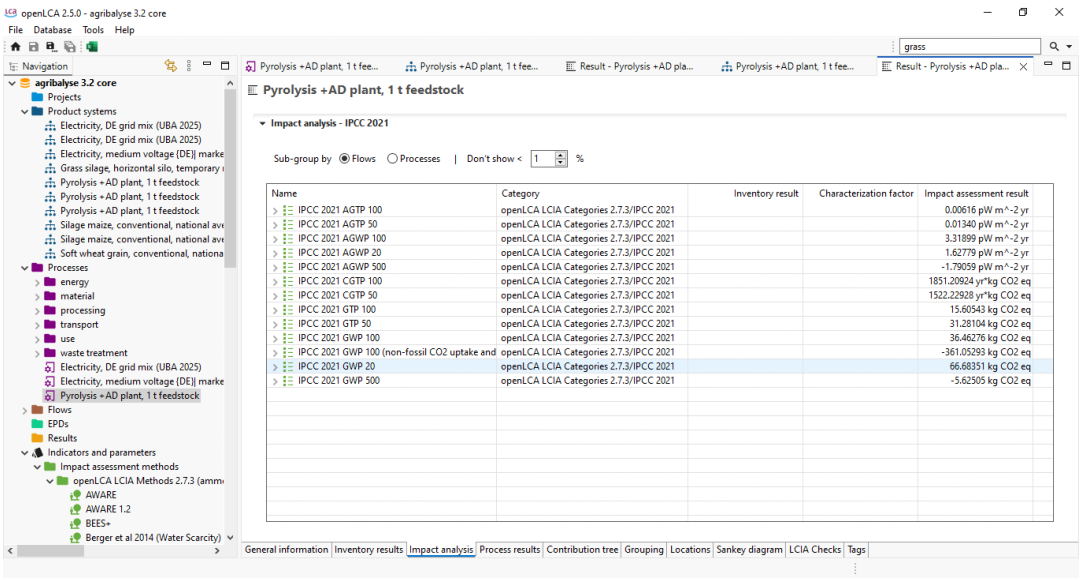


Figure 11: Impact analysis of the plant process: IPCC 2021 GWP 100 = 36.46276 kg CO2 eq per t of feedstock.

7.0 LCA results

The base case comes out at +36.5 kg CO2e per tonne of feedstock. The plant causes 119.3 kg CO2e, and the exported electricity avoids 82.9 kg of that, about 69 %. Wheat is the largest single source at 37.4 kg (31 % of the burden), followed by N2O from the digestate at 20.8 kg. Figure 12 shows the steps.

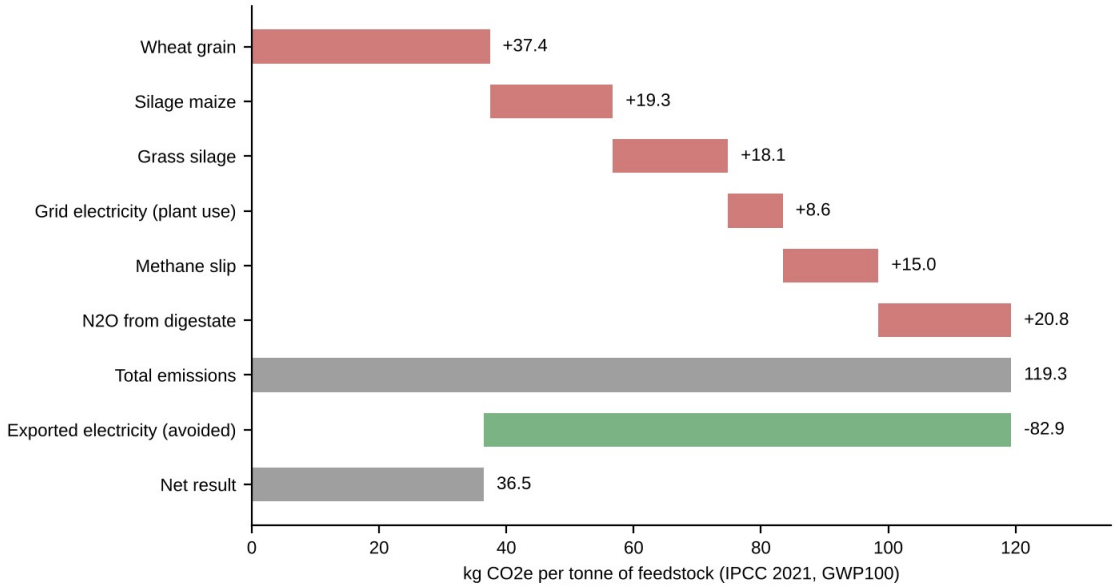


Figure 12: Contributions to the GWP per tonne of feedstock (base case). Red adds emissions, green avoids them, grey shows totals.

7.1 Scenario: heat used by a customer

If the exported heat (156.6 kWh per t) replaces heat from a gas boiler (240 g CO2e per kWh of gas at 90 % efficiency, which is 0.267 kg CO2e per kWh of heat), the result turns negative. I do not claim this credit in the base case, because I am not assuming that anyone buys the heat.

Case	GWP (kg CO2e per t feedstock)
Base case: no heat credit	+36.5
Heat used: credit of 41.8 kg CO2e	-5.3

Table 5: Heat scenario.

7.2 Sensitivity to the methane slip

Methane slip is the least certain of the direct emissions. Every percentage point adds 7.9 kg CO2e per tonne.

Methane slip	GWP (kg CO2e per t feedstock)
1.0 %	+29.3
1.9 % (base case)	+36.5
3.0 %	+45.2

Table 6: Sensitivity to the methane slip.

7.3 Checking Excel against openLCA

I calculated the same inventory separately in Excel and in openLCA. The two results agree to the third decimal, so amounts, units, signs and factors were transferred without errors. The difference of 0.004 kg comes from rounding of the inputs.

Model	Result (kg CO2e per t feedstock)
Excel (inventory x GWP factors)	36.459
openLCA (product system, IPCC 2021 GWP100)	36.463

Table 7: Cross-check of the two models.

8.0 Life cycle costing

The cost model reads the same inventory rows as the GWP. Each flow in Table 4 is multiplied by a price, and the plant costs (capital, maintenance, staff) are added on top. Table 8 lists the inputs and where each one comes from.

Input	Value	Source
Lifetime	20 years	Balussou et al. (2012); DBFZ Report 28
Discount rate	6 %	Balussou et al. (2012). DBFZ Report 28 uses 2.8 to 6.5 %
CAPEX	4,000 EUR per kWel, about EUR 26.3M for 6.6 MWel	Balussou et al. (2012): a 500 kW plant from 2012, so a conservative figure for this size
Maintenance, insurance, administration	2 % of CAPEX per year	DBFZ Report 28 (1 % + 0.5 % + 0.5 %)
Wage	EUR 40 per hour	topagrar; Destatis gives EUR 43.40 per hour as the 2024 average labour cost
Staff hours	20,000 h per year	3.0 h per kWel and year. FNR (2008) gives 3 to 7 h, so this is the low end for a large plant
Silage maize	EUR 40 per t	topagrar 2025 (EUR 39.4 to 44.8 per t)
Feed wheat	EUR 183.7 per t	Sachsen Agrarmarkt, 3 November 2026
Grass silage	EUR 40 per t	LfL Bayern. The costs are from 2006/07, so they are outdated
Grid electricity bought	EUR 0.178 per kWh	BDEW Strompreisanalyse, October 2025
Electricity sold	EUR 0.089 per kWh (spot), EUR 0.1811 per kWh (EEG case)	Bundesnetzagentur: 2025 day-ahead average; average award of the October 2025 biomass tender
Biochar	EUR 400 per t	Low end of EUR 300 to 2,000 per t reported for Europe (Farmers Journal, 2023)
Heat (heat scenario only)	EUR 0.04 per kWh	Inside the 2 to 8 ct per kWh range quoted by topagrar

Table 8: Cost inputs and their sources.

Manure and green waste are at zero cost, with no gate fee, and I give no fertiliser value to the digestate. Both are my own choices.

Table 9 shows the results at 80 % utilisation. The first two rows use the spot price for electricity, the last two the average award of the EEG biomass tender.

Case	Net cost (EUR per t)	NPV
Spot price, no heat sale	16.6	-EUR 44M
Spot price, heat sold	10.4	-EUR 28M
EEG tariff, no heat sale	-5.5	+EUR 15M
EEG tariff, heat sold	-11.8	+EUR 31M

Table 9: Cost results per tonne of feedstock (20 years, 6 %, 80 % utilisation).

At the spot price the plant does not cover its running costs. The operating cash flow is about -EUR 1.6M per year, and the capital cost comes on top, which gives the negative NPV. Utilisation matters less than the electricity price. At spot price and a 20-year life, the net cost is EUR 21.9 per t at 60 % utilisation, 16.6 at 80 % and 13.5 at 100 %.

The EEG cases are an upper bound. EEG support only covers electricity up to a limit (the Hoechstbemessungsleistung), which I did not model. The tender price is also not guaranteed, because it depends on winning a bid.

9.0 Assumptions and limitations

This is a screening-level LCA and LCC. The structure is sound, but a few inputs are my own assumptions, and the sourced ones carry real uncertainty. I would not quote the figures outside this report without saying so. Table 10 lists them.

Assumption or limit	Effect	Status and source
Methane slip 1.9 % of produced methane	About 4.0 kg CO2e per t for each 0.5 percentage point	Mean for CHP units (DBFZ / IEA Task 37). Whole agricultural plants range from 1.1 to 13.7 %, so this is a low value
Digestate nitrogen 6.0 kg N per t, 1 % N2O-N emission factor	Gives the +20.8 kg CO2e from N2O	Table value for liquid digestate (LfL Bayern), applied per t. The 1 % factor is the IPCC default
Pyrolysis share 10 % of the feedstock mass	Sets the biochar and pyrolysis energy	My assumption. It replaces the faulty sizing of the design report
Pyrolysis yields 30 / 40 / 30 %	Split of the pyrolysis energy	From the design report. Inside the slow-pyrolysis ranges in the literature
Bio-oil 17 MJ/kg	Part of the 153 kWh of pyrolysis energy	Plausible: the literature gives 16 to 26 MJ/kg (higher heating value)
Syngas 8 MJ/kg	Part of the 153 kWh of pyrolysis energy	My assumption. Pyrolysis gas is reported at 3 to 15 MJ/Nm³. My first value of 15 MJ/kg looked too high. Switching to 8 raised the result from +31.6 to +36.5 kg CO2e per t
Own electricity use 25 kWh per t (about 10 % of CHP electricity)	Gives the +8.6 kg CO2e from grid electricity	My assumption. Measured own use of 250 kW biogas plants is 5.5 to 8.5 %. I used more to cover pyrolysis and drying
35 % of CHP heat used on site; boiler efficiency 90 %	Only the heat scenario	My assumptions, no source found
French crop data for German crops	Crop GWP may differ in Germany	Proxy
Electricity factor is CO2 only (UBA 2025, estimate)	Slightly understates the avoided CH4 and N2O of the grid	Stated
No ecoinvent database	Fewer and older background data	Can be replaced later without changing the model structure
Biochar carbon storage and digestate value not credited in the LCA	The base case is conservative	Stated
Plant construction excluded from the LCA	Small share for long-lived plants	Stated
CAPEX from a small 2012 plant; grass silage cost from 2006/07	Cost results are rough	Stated
EEG cap not modelled	The EEG cases overstate revenue	To do

Table 10: Assumptions and limitations.

10.0 Conclusion

The plant concept as I modelled it causes about 36 kg CO2e per tonne of feedstock. Most of this comes from growing the crops and from the direct emissions of methane and nitrous oxide. The electricity that the CHP engine exports avoids about two thirds of it. If a customer buys the heat, the net result is slightly below zero.

On cost, the plant does not pay back at the 2025 spot price for electricity. It could pay back with an EEG tariff, but I have not modelled the limit on supported electricity, so that result is optimistic.

The results depend most on the pyrolysis assumptions, the methane slip and the electricity price. The next steps would be to find measured data for the pyrolysis products, to model the EEG cap, and to ask for access to ecoinvent so that the French proxy data can be replaced. The Excel model is built so that these changes can be made without changing its structure.

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